

Use the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology to analyze the Kaggle Heart Disease dataset. Follow and clearly explain each of the six phases: 1. Business Understanding, 2. Data Understanding, 3. Data Preparation, 4. Modeling, 5. Evaluation, and 6. Deployment. Act as an expert data scientist and a professor teaching a Master's-level Data Science course. The analysis should be clear, rigorous, and educational, explaining why important decisions are made rather than only showing code and outputs. Use standard Python libraries such as pandas, NumPy, matplotlib, seaborn, and scikit-learn. Include EDA, data cleaning and preprocessing, feature engineering and selection, multiple classification models and baselines, model evaluation, ensemble modeling, clustering, dimensionality reduction, association pattern mining, LSH/similarity search, outlier analysis, and a final evidence-based recommendation and conclusion.

ChatGPT: I inspected the uploaded Heart Disease CSV and treated the project as a complete CRISP-DM workflow rather than a collection of isolated modeling exercises. The original dataset contains 303 observations and 14 variables. The target is binary, there are no missing values, and an exact duplicate should be removed during data preparation.

Phase 1 - Business Understanding

Me: Define the business problem, objective, data-science objectives, and success criteria.

ChatGPT: The central problem is whether demographic, clinical, and physiological characteristics can be used to classify whether a patient has heart disease. This is a supervised binary-classification problem. Because a false negative represents a positive heart-disease case that was missed, recall is important. Accuracy alone is therefore insufficient; precision, F1-score, balanced accuracy, ROC-AUC, and confusion-matrix analysis should also be considered. Success criteria include substantially outperforming a simple baseline, achieving strong recall and ROC-AUC, maintaining reasonable precision, avoiding major overfitting, and remaining computationally practical and interpretable.

Phase 2 - Data Understanding

Me: Examine the structure and quality of the Heart Disease dataset and perform EDA.

ChatGPT: The dataset contains 303 original observations, 14 variables, 13 predictors, and one target variable. There are zero missing values and one exact duplicate, leaving approximately 302 unique observations after duplicate removal. The numerical variables are age, trestbps, chol, thalach, and oldpeak. Other variables are binary, categorical, or ordinal-coded values.

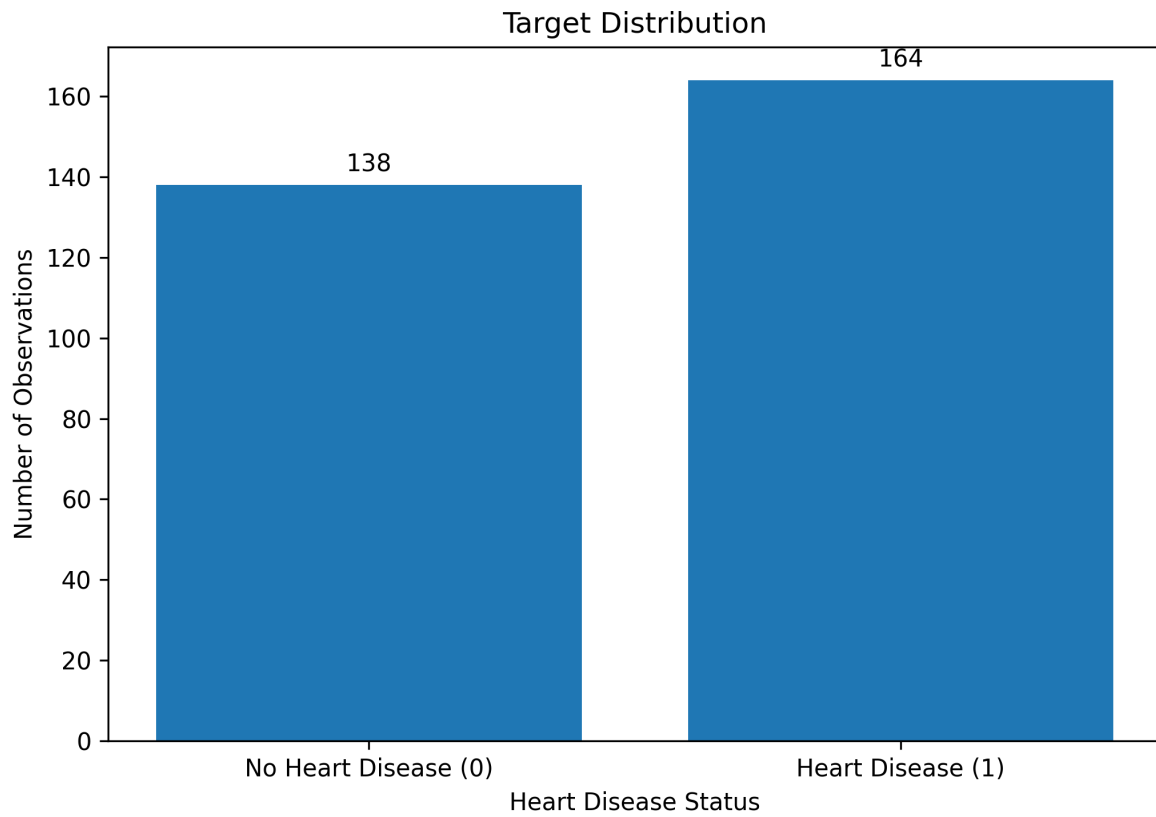


Figure: Target Distribution from the provided Heart Disease CSV

Interpretation: After duplicate removal, 138 observations are in the No Heart Disease class and 164 are in the Heart Disease class, approximately 45.7% and 54.3%.

Numerical Feature Distributions

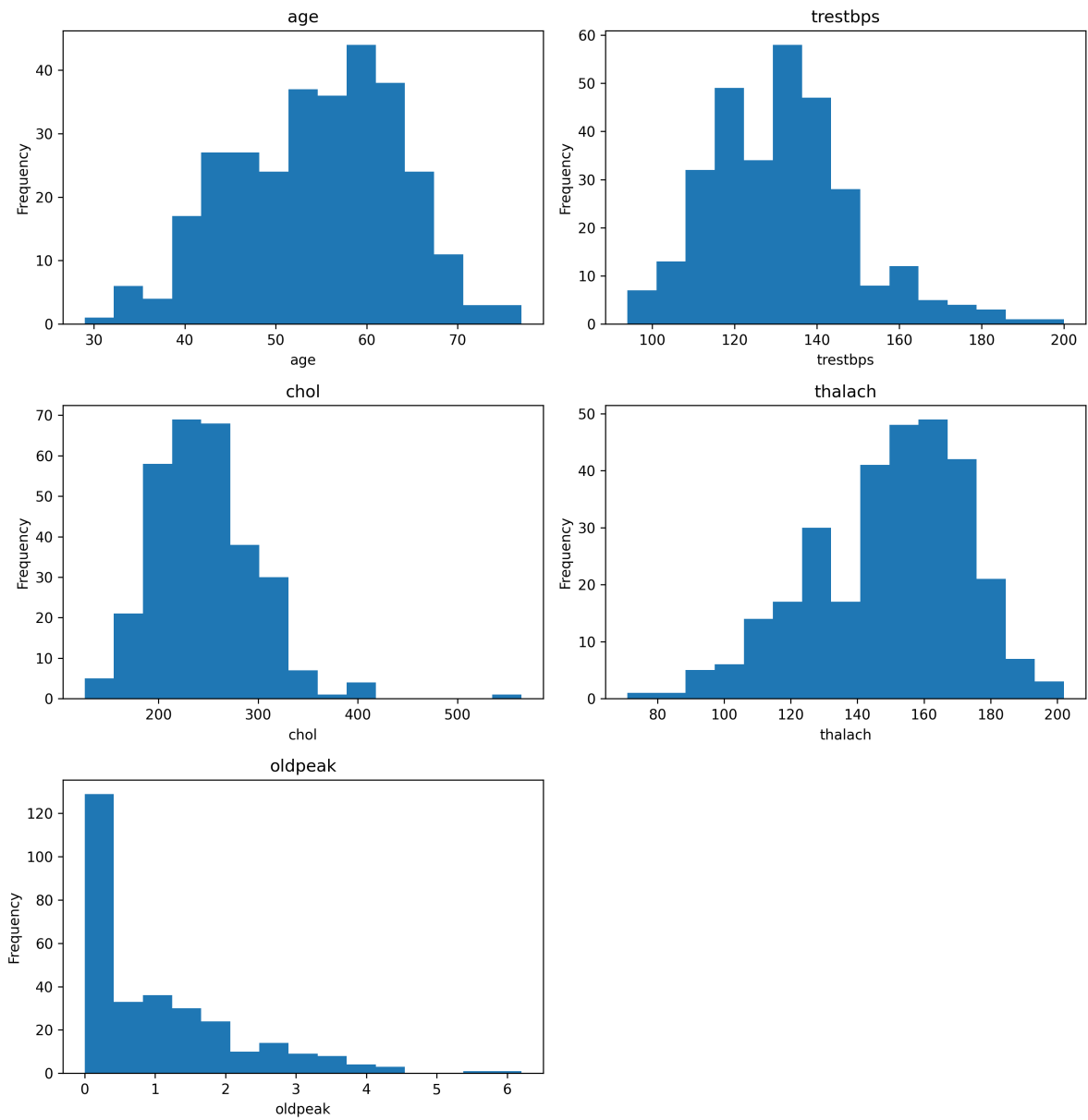


Figure: Numerical Feature Distributions

The histograms examine age, resting blood pressure, cholesterol, maximum heart rate, and ST depression.

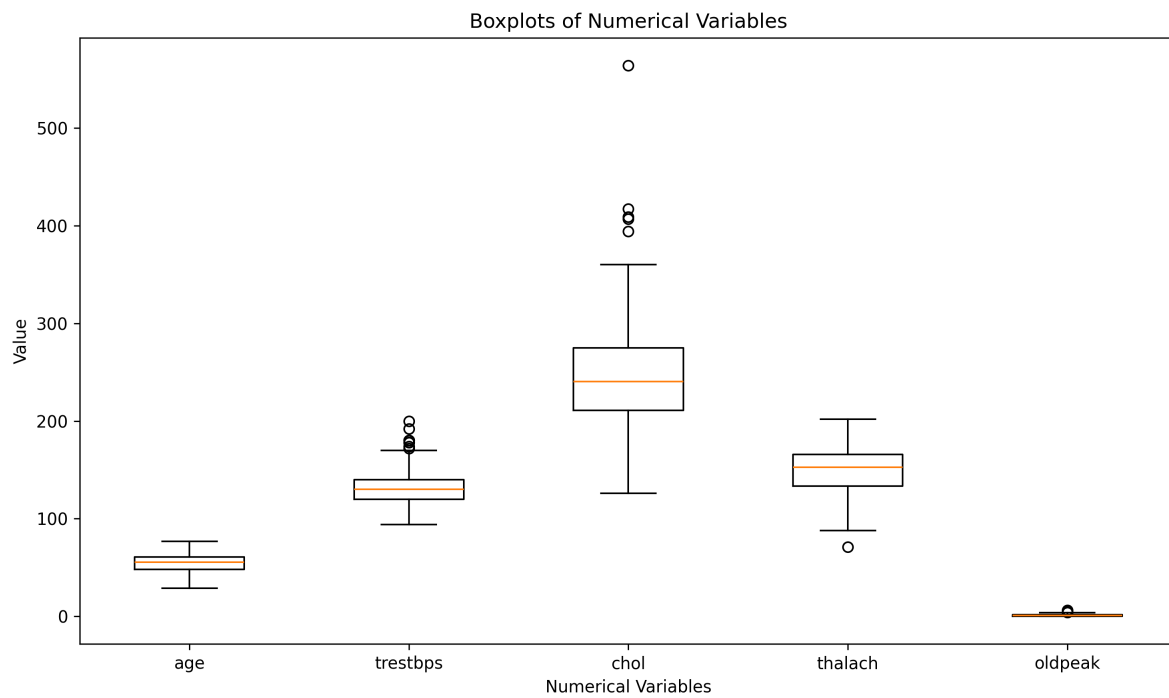


Figure: Boxplots of Numerical Variables

Potential extreme observations were identified, but unusual medical measurements are not automatically treated as errors.

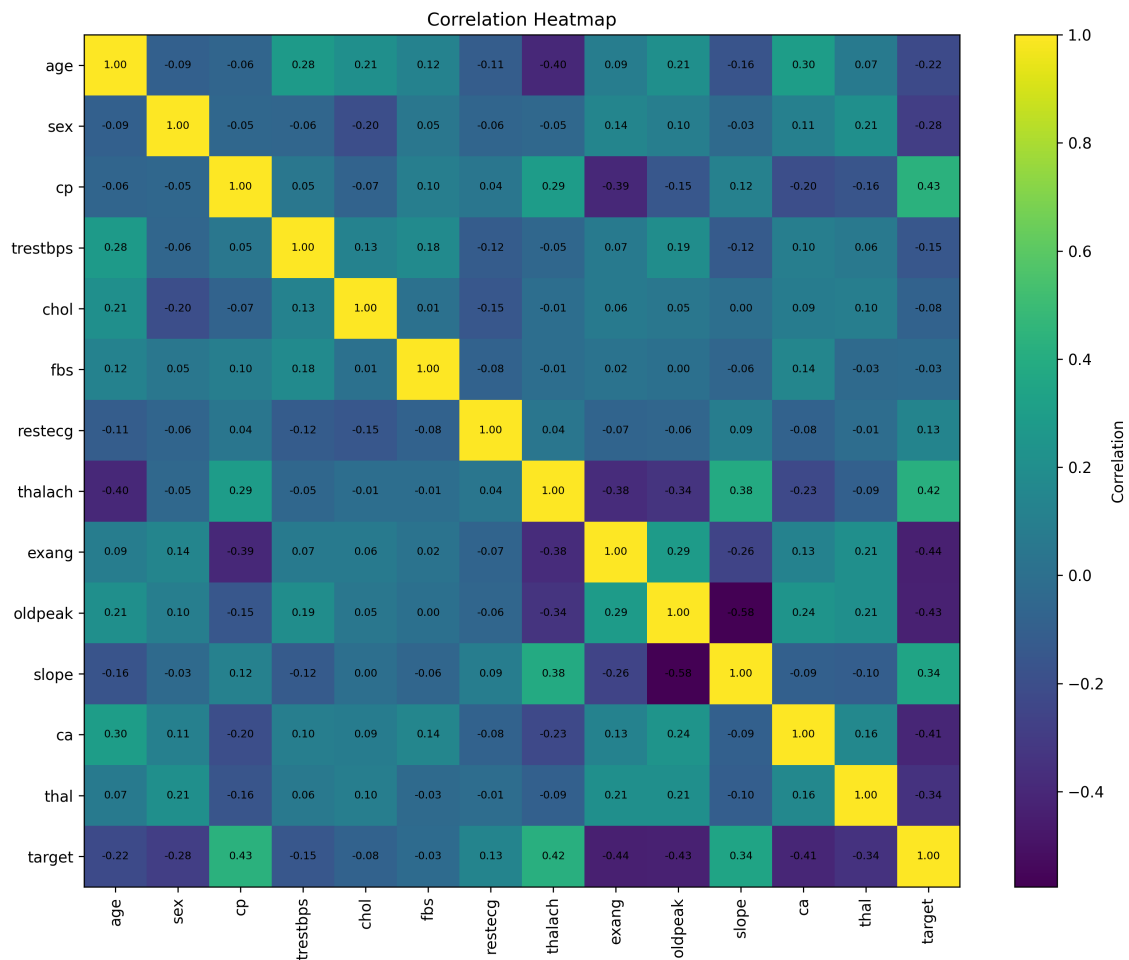


Figure: Correlation Heatmap

Correlation was used as an exploratory measure of linear association. Coded categorical variables require cautious interpretation.

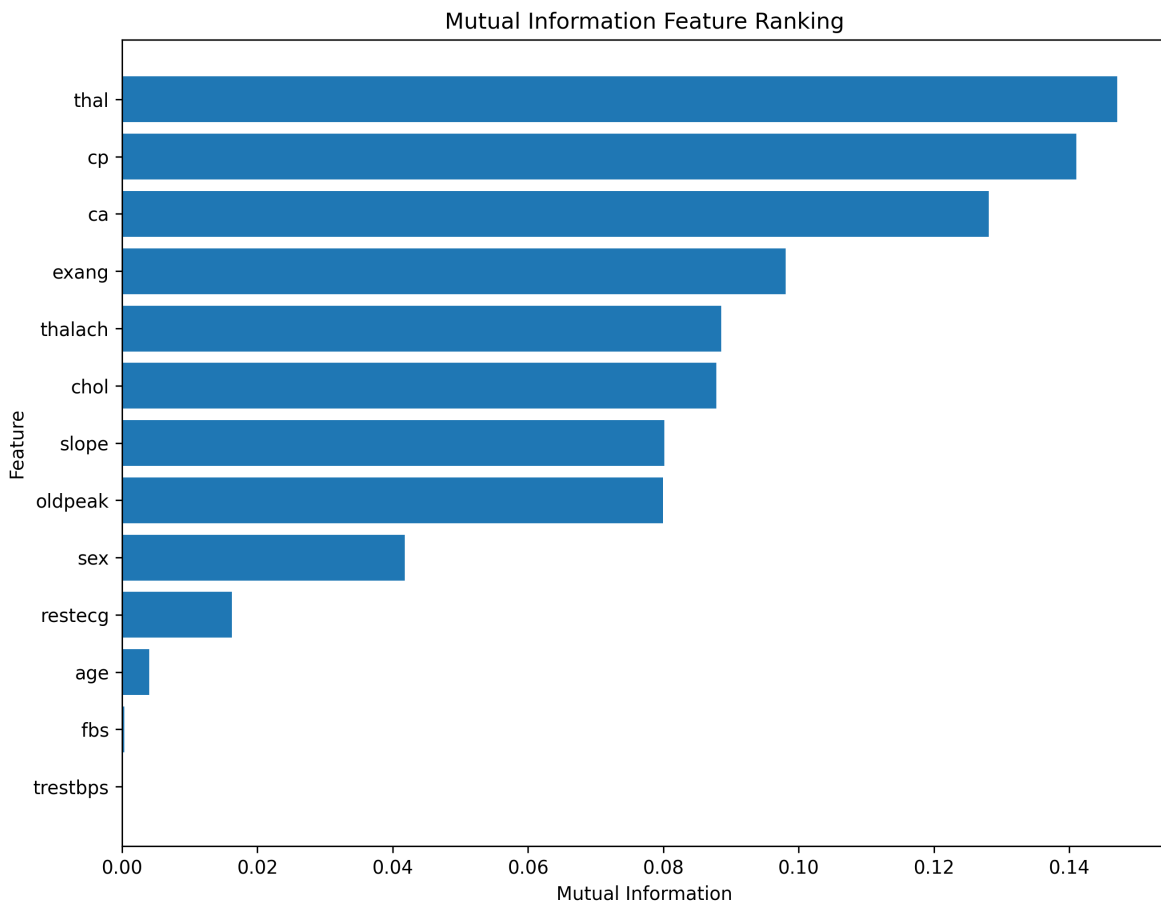


Figure: Mutual Information Feature Ranking

Mutual information was used as a complementary dependency measure. Variables including thal, cp, ca, exang, thalach, chol, slope, and oldpeak ranked among the more informative predictors.

Me: Summarize the main Data Understanding findings.

ChatGPT: The dataset is relatively small, contains no missing values, includes one exact duplicate, has a reasonably balanced target, and contains several potentially informative clinical variables. Potential outliers should be investigated rather than automatically removed.

Phase 3 - Data Preparation

Me: Explain cleaning, encoding, feature engineering, feature selection, and leakage prevention.

ChatGPT: The duplicate is removed. Numerical features are standardized with StandardScaler, while categorical and binary features are one-hot encoded with OneHotEncoder. ColumnTransformer and Pipeline keep preprocessing consistent and help ensure transformations are learned from the appropriate training data. A stratified 80/20 train-test split is used for holdout evaluation. Several domain-informed transformations were investigated, including age squared, blood-pressure-to-age ratio, cholesterol-to-age ratio, percentage of predicted maximum heart rate, an exercise-stress indicator, and a binary vessel-abnormality indicator. The documented experiment showed limited overall benefit from the added features, so the simpler representation was preferred when the added complexity did not provide a clear advantage. Feature selection used correlation, mutual information, model coefficients, permutation importance, and domain knowledge. Important variables repeatedly included ca, cp, thal, sex, slope, and oldpeak. Test data must not influence preprocessing or model training, so transformations are fit only on the appropriate training data through the pipeline.

Phase 4 - Modeling

Me: Establish a baseline and compare classification algorithms.

ChatGPT: A majority-class classifier serves as the naïve baseline. Its documented accuracy is approximately 54.4% with ROC-AUC 0.50. It establishes a simple reference point that learned models should exceed. The classification models include Logistic Regression, KNN, SVM, Decision Tree, Random Forest, Extra Trees, and Gradient Boosting. Logistic Regression is attractive because the target is binary, the dataset is small, computation is efficient, and the coefficients are interpretable. KNN and SVM offer alternative approaches, while tree models provide nonlinear modeling and interaction effects. Soft Voting and Stacking were also evaluated. The documented Stacking result was approximately 85.1% accuracy, 88.5% recall, 86.7% F1-score, and 91.1% ROC-AUC. K-means clustering, PCA, association-rule mining, LSH similarity search, and outlier analysis provided additional perspectives.

Phase 5 - Evaluation

Me: Perform rigorous evaluation using cross-validation and a holdout set.

ChatGPT: Two complementary evaluation strategies were used: stratified 10-fold cross-validation and a stratified 80/20 holdout test set. The primary metrics are accuracy, balanced accuracy, precision, recall, F1-score, and ROC-AUC. Cross-validation averages performance over multiple stratified folds, while the holdout evaluation measures performance on a reserved test set.

Model Comparison — Stratified 10-Fold Cross-Validation

Model	Accuracy	Recall	F1-Score	ROC-AUC
Majority Baseline	54.4%	100.0%	70.4%	50.0%
Logistic Regression	83.8%	86.2%	85.3%	90.8%
KNN	85.0%	87.0%	86.4%	89.6%
SVM	83.8%	85.4%	85.1%	90.8%
Decision Tree	78.4%	77.0%	79.2%	80.8%
Random Forest	83.4%	84.6%	84.5%	90.1%
Extra Trees	83.4%	86.9%	85.1%	89.3%
Gradient Boosting	79.7%	81.6%	80.9%	86.4%

Figure: Model Comparison - Stratified 10-Fold Cross-Validation

Model	Accuracy	Recall	F1-Score	ROC-AUC
Majority Baseline	54.4%	100.0%	70.4%	0.50
Logistic Regression	83.8%	86.2%	85.3%	90.8%
KNN	85.0%	87.0%	86.4%	89.6%
SVM	83.8%	85.4%	85.1%	90.8%
Decision Tree	78.4%	77.0%	79.2%	80.8%
Random Forest	83.4%	84.6%	84.5%	90.1%
Extra Trees	83.4%	86.9%	85.1%	89.3%
Gradient Boosting	79.7%	81.6%	80.9%	86.4%

Interpretation: KNN achieved the highest accuracy and F1-score in this comparison, while Extra Trees achieved the highest recall. Logistic Regression and SVM achieved the highest ROC-AUC. Logistic Regression was selected because it provided a strong overall balance of performance, interpretability, computational efficiency, and simplicity.

Me: Explain the holdout results and confusion matrix.

ChatGPT: For the actual CSV holdout run, Logistic Regression achieved approximately 85.25% accuracy, 85.29% precision, 87.88% recall, 86.57% F1-score, and 89.72% ROC-AUC.

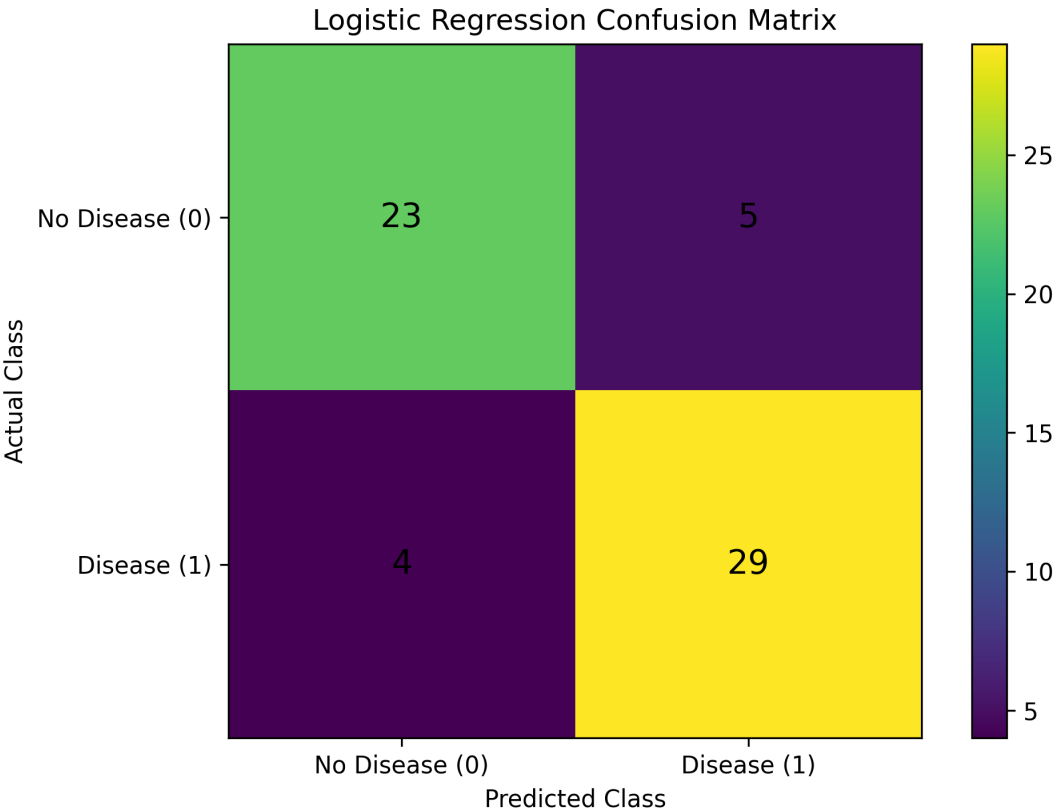


Figure: Logistic Regression Confusion Matrix

ChatGPT: The holdout confusion matrix contains 29 true positives, 23 true negatives, 5 false positives, and 4 false negatives. The false negatives are important because they represent positive cases the model failed to identify.

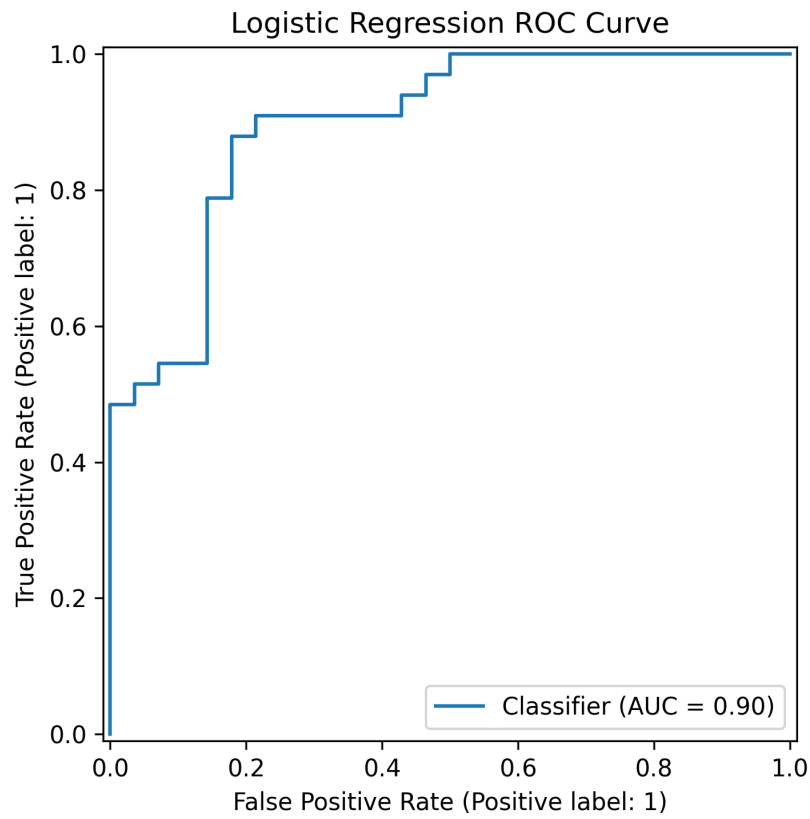


Figure: Logistic Regression ROC Curve

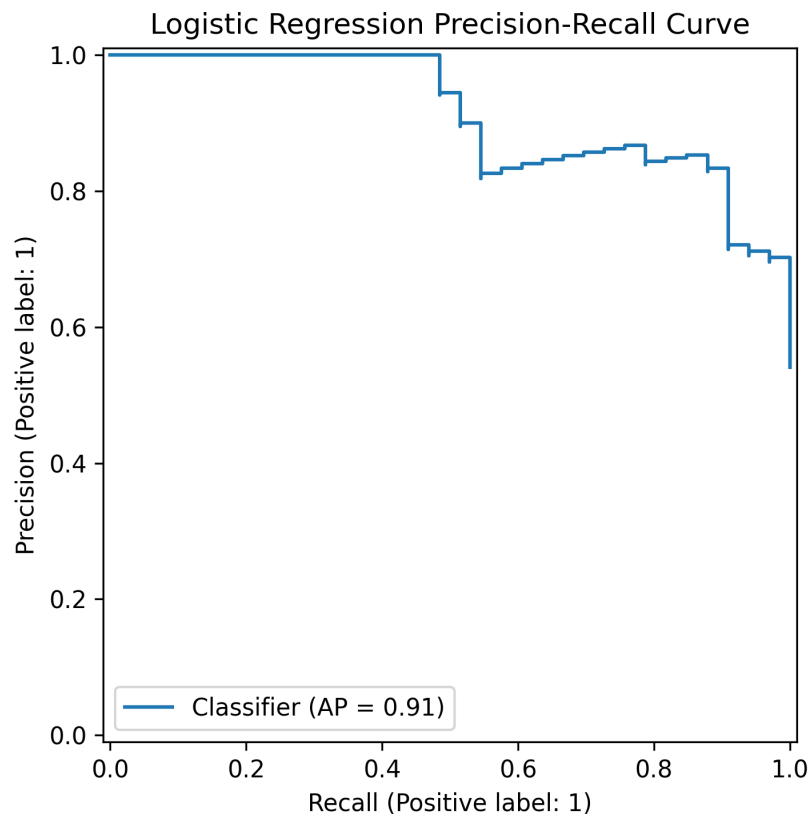


Figure: Logistic Regression Precision-Recall Curve

ChatGPT: The ROC-AUC of approximately 0.897 indicates strong discrimination within this dataset, while not establishing clinical validity. The Precision-Recall curve shows the trade-off between precision and recall as the classification threshold changes.

Phase 6 - Deployment

Me: Explain how the selected model could be deployed.

ChatGPT: The selected Logistic Regression model can be implemented as a reproducible prediction pipeline. Patient data would be validated, numerical and categorical features transformed, a probability generated, and a classification threshold used to produce the predicted heart-disease class. For this assignment, deployment is an educational/research prototype rather than a clinical diagnostic system. A real deployment would require monitoring data quality, model performance, calibration, data drift, threshold performance, and subgroup/fairness performance, along with independent validation and clinical review.

Final Recommendation and Conclusion

Me: Provide the final recommendation and summarize what I should understand from the analysis.

ChatGPT: Logistic Regression is recommended as the primary model. KNN achieved slightly higher accuracy and F1-score and Extra Trees achieved slightly higher recall, but Logistic Regression offered a strong overall balance of predictive performance, ROC-AUC, interpretability, computational efficiency, and simplicity. The main lesson is that the most complex model is not automatically the best. The analysis also shows why outliers should not automatically be removed, why categorical codes require careful treatment, and why preprocessing must avoid test-data leakage. The dataset is small, has limited population representation, and lacks independent external validation. Overall, Logistic Regression is a practical research model for this dataset, but much more validation is needed before any real healthcare use.